

Unraveling the Causal Relationship Between Economic Policy Uncertainty and Exchange Market Pressure in BRIC Countries: Evidence From Bootstrap Panel Granger Causality

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Abstract

This study examines the causal relationships between exchange market pressure and both global and domestic economic policy uncertainty among the Brazil, Russia, India, and China economies. We employed the bootstrap panel Granger causality approach, having found cross-sectional dependency among the countries as an indicator of their mutual developmental characteristics. The individual country analysis shows that, between exchange market pressure and global economic policy uncertainty, only China has bidirectional causality, while no causality was found for Russia. Between exchange market pressure and domestic economic policy uncertainty, India and Russia have bidirectional causality. Domestic economic policy uncertainty does not matter for China's exchange market pressure; therefore, China has one-way causality from exchange market pressure to domestic economic policy uncertainty. Overall, in these countries, exchange market pressure is influenced by global economic policy uncertainty while both exchange market pressure and domestic economic policy uncertainty mutually interact to influence each other.

Keywords

exchange market pressure, global economic policy uncertainty, domestic economic policy uncertainty, BRIC

Introduction

The world is now characterized by increasing economic policy uncertainty (EPU) because of the growing interrelationships among many economies. Sudden change or expected changes in macroeconomic policies, both home and abroad, often generate disruption in macroeconomic activities, thereby causing delays in decision-making and increasing risk. Both domestic and foreign decision-making could become more complex when the direction of such macroeconomic policies is unknown. Historical evidence shows that EPU could have both domestic and foreign origins, especially when it is coming from a large economy which is big enough to impact the rest of the world. It has become important to understand the true origin of uncertainty spillover into the macro-economy. Thus, the overall aim of this study is to examine the causal interactions between exchange market pressure (EMP) and both domestic and global EPU in BRIC countries (Brazil, Russia, India, and China).

Prior to the global financial crisis of 2008-2009, which led to heightened levels of macroeconomic policy

uncertainty, previous financial crises (such as the Latin American debt crisis of 1982, the stock market crisis of 1987 and the Asian crisis of 1997-1998) have proven that the world's economies are interconnected in a way, because crisis emanating from a large economy has spread into other economies. In a defensive response, economic policies from large economies often further amplify the impact leading to an enormous global economic policy uncertainty (GEPU; Baker, Bloom, & Davis, 2016; Bloom, 2017; International Monetary Fund [IMF], 2014; Luk, Cheng, Ng, & Wong, 2017; Yu, Fang, & Sun, 2018).

After the global financial crisis led by the United States, the world has witnessed other incidents that have caused an upsurge of EPU; for example, uncertainties surrounding the

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European banking crisis, Chinese leadership transition, and the U.S. fiscal policy between 2012 and 2013. Recent examples are the uncertainties surrounding Trump's election in the United States and Brexit in Europe in 2016. The most recent issue of concern is the protectionist trade policy coming from the United States, which has put the emerging markets' economies under pressure. There have been constant threats to global trade following the trade and currency war threats by the United States to most large emerging market economies, especially China. There are two major concerns about this: First, if the trade and currency wars continue and extend to other economies, there arises uncertainty about the BRIC's intercontinental trade and this threatens the realization of their projected economic growths. Second, the impacts of their policy response have influence on their domestic economies and the potential to spread across other countries, especially to other trade partners. Policy debates about the domestic and global spillover effects of policy uncertainty from advanced economies continue to arise because the rising trend of economic alliance among large emerging economies is being threatened by potential economic disruption (IMF, 2013).

As relations among the economies of many nations get deeper, foreign transactions get more complex and international spillovers of economic uncertainty are inevitable. Among the various spillover dimensions that the effects of economic crises can take, the alteration of the exchange market is principal because of its high liquidity. Other channels include policy interest rate, long-term interest rates, international bank lending, and both short- and long-term portfolio flows (Aizenman & Binici, 2016). The foreign exchange rate becomes susceptible to GEPV via trade and financial openness, free capital mobility, and operation of large financial and external sectors (Georgiadis, 2016; IMF, 2013; Luk et al., 2017).

In a financially open economy with free capital mobility, the foreign exchange market is prone to high volatility of portfolio flows and sudden stops in capital flows. Trade openness puts the exchange market through the pressure of trade partners' economic disruption and external disturbances. The presence of an external sector in a home country means that the country may not be shielded from unpredictable events arising from trading partners, such as price shocks, a sudden rise in interest rates, or a sharp slowdown in their growth, causing a sharp cut in expectations and returns. Management policies and the exchange rate regime determine the role of the exchange market in such international spillovers, but large fluctuations in exchange rate are often nonnegligible. The effects include subversion of the balance sheet effects, external imbalances, and financial instability. Significant fluctuations in the exchange rate are, therefore, critical for policy designs, especially for export-led economies (Aizenman & Binici, 2016).

In countries operating free floating exchange rate regimes, all the pressure in the foreign exchange market is felt through

exchange rate fluctuations, but in countries operating fixed exchange rate regimes, all the foreign exchange market pressure is felt through changes in the reserves. The central bank intervenes through the foreign reserves to stabilize the exchange rate such that pressures in the exchange rate are not observed, but are, instead, reflected through changes in reserves. However, for countries operating intermediate regimes, the foreign exchange market pressure is felt partly through exchange rate fluctuations and partly through changes in reserves. We, thus, consider all plausible changes in both exchange rates and foreign reserves as captured by the EMP. The EMP is the most suitable means of measuring total pressure in the foreign exchange market because it simultaneously captures both exchange rate and foreign reserve changes.

The theoretical explanations about how GEPV or domestic economic policy uncertainty (DEPU) is connected with EMP lead to three different types of relationships: First is the direct relationship which comes via hedging motive. Intuitively, investors tend to invest in safer currencies during uncertainty. Thus, if a currency is relatively safer during high uncertainty, it will appreciate rather than depreciate, but unsafe currencies tend to witness depreciation pressure during high uncertainty. In addition, if domestic uncertainty rises above global uncertainty, then foreign currency denominated investment will be more attractive to domestic investors at that particular time. This implies that domestic currency will be under pressure for devaluation or depreciation relative to the foreign currency, thereby causing movements in the EMP (Balcilar, Gupta, Kyei, & Wohar, 2016; Benigno, Benigno, & Nisticò, 2012). Second is the indirect relationship which lies in the knowledge that sudden movements in the EMP components (exchange rates and foreign reserves) are due to unanticipated changes in macroeconomic fundamentals, such as output, domestic and foreign investments, interest rates, and trade. As long as these macroeconomic fundamentals are subject to actual or expected fluctuations caused by global or domestic uncertainty, they are indirect transmission channels through which uncertainty is connected to EMP (Balcilar et al., 2016). Third, inherent or expected policy response to macroeconomic volatilities suggests that pressures in the exchange market will affect the level of uncertainty. As exchange rate volatility influences demand and prices, it prompts macroeconomic policy changes, oftentimes through interest rates manipulation. On these bases, we test the related hypotheses as follows:

Null Hypothesis (H0): There is no significant causal relationship between GEPV and EMP.

Hypothesis 1 (H1): There is a significant causal relationship between GEPV and EMP.

Null Hypothesis (H0): There is no significant causal relationship between DEPU and EMP.

Hypothesis 1 (H1): There is a significant causal relationship between DEPU and EMP.

The implications of this study are for policymakers to note: First, the role of global economic changes in explaining the foreign exchange market activities, and consider that the reactive policy response will determine the severity of global uncertainty effects on the foreign exchange; second, that DEPU and EMP mutually interact to reinforce each other; therefore, soothing monetary policy might ease the pressure on both sides; third, that the mutual economic characteristics shared among countries suggest that external shocks affecting one country will likely be transmitted through the exchange markets into the domestic sector; fourth, that gains from openness and setbacks from negative effects of global uncertainty shocks are tradeoffs which require careful considerations while taking economic policy measures.

Our study contributes to literature in a threefold manner: First, we analyze the causality between EMP and GEPU for each of four countries, and do same for DEPU. This has not been investigated in previous literature.

Second, we apply the bootstrap causality procedure, which gives reliable results in the presence of slope heterogeneity and cross-sectional dependency. Accounting for cross-sectional dependency is crucial for what it shows about the economic alliance, mutual growth and developmental characteristics among our sample countries.

Finally, we show each country's causal relationships between EMP and GEPU, and between EMP and DEPU. Most importantly, we found that both GEPU and DEPU Granger cause EMP, and DEPU has a feedback relation with the EMP in the panel of countries.

The rest of this study is organized as follows: next, we discuss BRIC economies in brief, review relevant literature in the "Literature Review" section, present the econometric models and data used in our analysis in the "Data and Methodology" section, interpret the results obtained in the "Empirical Results" section, present policy implications of our findings in the "Policy Implications" section, and finally present conclusions and recommendations in the "Conclusion" section.

The BRIC Economies in Brief

The BRIC countries were preferred for this study because of the growing importance of their market-oriented economies and economic alliance with developed markets. They jointly account for the largest share of global economic growth and the world's growing consumption. Their extraordinary strides into development and vibrant financial markets attract a lot of global attention, thereby attracting high capital inflow from the rest of the world. They also operate large external sectors and their economies thrive on exports, thus they are becoming dominant in international trade and engage in considerable foreign exchange via trade of strategic commodities. Brazil, for instance, is a major supplier of raw materials, semi-manufactured and manufactured products, Russia is a major oil exporter, China takes the lead in manufacturing,

while India is at the fore of the world's food production and services.

The global economic focus is gradually shifting to BRIC, as they have become important markets to the developed countries, both as major trade partners and as competitors (Boyer & Truman, 2005). It was projected that these four will overtake the largest Western economies by 2039 (Goldman & Sachs as cited in Wilson & Purushothaman, 2003).

Their currencies have become prominent in international exchange by interacting with the strongest currencies of the world. Thus, exchange rate variability is an issue of concern for BRIC because of the involvement of their local currencies in bilateral trade, especially China and Russia whose currencies have been involved for about a decade. Although this development is expected to reduce transaction costs among the BRIC countries and enhance their political strength to achieve favorable international negotiations, it would, most importantly, allow BRIC to diversify their foreign reserves so as to manage other risks. However, as long as the bilateral trade between United States and European countries remains stronger than the trade among BRIC, they are obliged to use the USD in most of their transactions (Maradiaga, Zapata, & Pujula, 2012).

The global uncertainty shocks that hit their currencies through their interaction with stronger currencies are worth checking. This is noteworthy because pressures on their currencies have huge effects on their net worth in real terms, as shown through the balance sheet. Also, their investors take decisions daily based on the value and volatility of their currencies. Furthermore, as emerging economies, their domestic price levels are highly sensitive to their exchange rate fluctuations, thereby aggravating inflationary pressures in the domestic economy (Calvo & Reinhart, 2002). It is, therefore, timely to investigate possible influence of global shocks on their EMP.

Literature Review

Theory suggests that uncertainty may dampen economic activity. Literature providing empirical evidence of this includes Kang and Ratti (2013); Handley (2014); Kang, Lee, and Ratti (2014); Leippold and Matthys (2015); and Bernal, Gnabo, and Guilmin (2016). The impact of external uncertainty on economic activities has been documented on consumer prices, equity prices, interest rates and services by Belke and Osowski (2017) and Yu et al. (2018). Furthermore, Georgiadis (2016) found that external uncertainty will spill over into domestic activities via trade openness, the exchange rate regime and financial market development, among other factors. Specifically, Krol (2014); Kido (2016); Arbatli, Davis, Ito, Miake, and Saito (2017); and Beckmann and Czudaj (2017) established that EPU does influence exchange rates.

There are also studies providing insight into the spillover effects of international EPU into other countries. For

instance, Ko and Lee (2015); Bouri, Gupta, Hosseini, and Lau (2017); and Das and Kumar (2018) examined the spillover of international uncertainty into stock prices. Available literature further shows that there are international spillovers of uncertainty from larger economies to smaller economies (see Belke & Osowski, 2017; Bernal et al., 2016; Colombo, 2013; Dakhlaoui & Aloui, 2016; Luk et al., 2017; Sarwar, 2012). However, none has shown the linkage between EPU (global or domestic) and EMP.

The idea of EMP was originally proposed by Girton and Roper (1977) who analyzed shocks to foreign exchange in a model that explains both exchange rate movements and official reserves intervention. Weymark (1995, 1997, 1998) further applied the model consistent formula by adding the observed changes in exchange rate and the avoided change in exchange rate through intervention. This validates the pioneer measurement of EMP. Eichengreen, Rose, and Wyplosz (1996) normalized and weighted the components of the index with their volatilities to derive the EMP. These measures of EMP have motivated the identification of some of its external and domestic determinants.

Aizenman and Hutchison (2012) revealed that emerging markets with higher total foreign liabilities are at a higher risk of global financial crisis. The external factors found to drive EMP include volatile capital inflows such as non-FDI inflows, FDI inflows, capital controls, and interest rate differential (see also Aizenman & Binici, 2016; Aizenman & Hutchison, 2012; Feldkircher, Horvath, & Rusnak, 2014; Gochoco-Bautista & Bautista, 2005; Hegerty, 2012).

Exchange market pressure is also driven by domestic factors, which include price stability, domestic savings, domestic credit growth, devaluation expectations, monetary policy, and fiscal deficit (Akram & Byrne, 2015; Bird & Mandilaras, 2006; Feldkircher et al., 2014; Gochoco-Bautista & Bautista, 2005; Hallwood & Marsh, 2004; Hegerty, 2012; Panday, 2015). Sequel to all these determinants of the EMP in the literature, it is considered as a key variable in international finance. None of these are known to have addressed the interaction between EPU (global or domestic) and EMP.

Data and Methodology

To ascertain the causal relationship between GEPU and the EMP of the BRIC economies, we use monthly time series data from 1995 M01 to 2018 M02 for each country. The indices of DEPU and GEPU developed by Baker et al. (2016) were obtained from <http://www.policyuncertainty.com/>

The national monthly EPU index is constructed from a search of daily newspaper archives on articles related to a set of three terms, namely “economy,” “policy,” and “uncertain.” The relative frequency of EPU is the share of each country’s newspaper articles that focus on these three terms in a particular month. We employ these historical data for Brazil, China, India, and Russia as a measure of the degrees of domestic EPU in their respective economies.

The GEPU index is the addition of 20 countries’ EPU indices weighted by their relative shares of current-price GDP. These 20 countries (Australia, Brazil, Canada, Chile, China, France, Germany, Greece, India, Ireland, Italy, Japan, Mexico, the Netherlands, Russia, South Korea, Spain, Sweden, the United Kingdom, and the United States) jointly account for about 70% of the world’s output. Davis (2016) compiled a GDP-weighted average of their EPU indices for each month in three major steps: First, each national EPU index was renormalized to a mean of 100; second, to generate a balanced panel of EPU index, a regression-based method was employed in assigning missing values to affected countries; third, the GDP data derived from IMF’s World Economic Outlook Database were used to compute the GDP-weighted average of the EPU indices, which gives the monthly GEPU index.

Foreign exchange rate and international reserves minus gold were used to construct the EMP. We obtained data from the IMF and International Financial Statistics (IFS) database. To calculate EMP, we use the definition of EMP by Aizenman and Pasricha (2012) that

$$EMP_t = \left(\frac{e_t - e_{t-1}}{e_t} - \frac{ir_t - ir_{t-1}}{ir_t} \right) \times 100, \quad (1)$$

where e_t denotes the nominal exchange rate, national currency per U.S. dollar, ir_t denotes international reserves (minus gold) in U.S. dollars, while t denotes time. Thus, an increase e_t implies increase in EMP while an increase in ir_t implies a decrease in EMP.

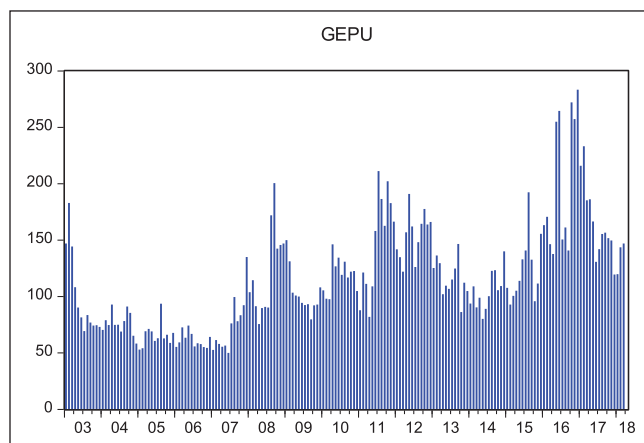
Table 1 presents the summary statistics for EMP and DEPU for an individual country. It also shows the summary statistics for the GEPU index. The reports show that India has the lowest DEPU with the lowest mean value of 95.532, while China has the highest DEPU with the highest mean value of 159.936 and this value is closely followed by Brazil’s mean value of 158.914. Again, considering the absolute values of EMP, China takes the lead, with the highest mean value of -0.014 , also followed closely by Brazil’s mean, while India has the lowest mean value of -0.008 in this case. China has the highest variation in DEPU with a standard deviation value of 115.482 while India has the lowest variation with a standard deviation value of 52.639. In terms of EMP, Russia has the highest variation with a value of 0.063 while China has the least variation with a value of 0.021. The standard deviation value of 46.897 recorded under the GEPU shows that its variance is not as high as DEPU in any of the BRIC countries. The mean value of 116.325 shows that GEPU is relatively low compared with the DEPU of most the BRIC countries. From the Jarque–Bera test, the null of normality is rejected at 1% significance level, with the evidence of fat tails; all the variables for each country have excess kurtosis and are skewed to the right.

Table 1. Summary Statistics for Each Country's Variables.

	<i>M</i>	Median	Maximum	Minimum	<i>SD</i>	Skewness	Kurtosis	Jarque–Bera
GEPU	116.325	107.941	283.319	50.116	46.897	1.040	4.258	45.316*** (0.000)
DEPU								
Brazil	158.914	133.500	676.955	22.296	99.378	1.928	8.029	307.888*** (0.000)
China	159.936	123.140	694.849	26.1441	115.482	1.817	7.1732	234.754*** (0.000)
India	95.532	82.403	283.689	24.940	52.639	1.214	4.359	59.361*** (0.000)
Russia	138.133	116.717	400.017	24.111	76.118	1.080	3.752	40.085*** (0.000)
EMP								
Brazil	−0.013	−0.013	0.230	−0.133	0.056	0.746	5.560	67.367*** (0.000)
China	−0.014	−0.017	0.056	−0.059	0.021	0.552	3.607	12.163*** (0.000)
India	−0.008	−0.010	0.200	−0.089	0.036	1.196	8.318	260.655*** (0.000)
Russia	−0.009	−0.014	0.273	−0.135	0.063	1.142	5.817	100.819*** (0.000)

Note. The *p* values for Jarque-Bera test are in parentheses. GEPU = global economic policy uncertainty; DEPU = domestic economic policy uncertainty; EMP = exchange market pressure.

*** indicates the rejection of the null of normality at 1% level of significance.

**Figure 1.** The GEPU index.

Note. GEPU = global economic policy uncertainty.

The graphic illustrations of these indices are presented in Figures 1 to 3. The evolution of the GEPU index in Figure 1 has some significant spikes indicating important highpoints of global uncertainty. The first highpoint is the seen around 2003 as an effect of the Gulf War II, followed by the peak around 2008, signifying the global financial crisis. Another highpoint is observed between 2011 and 2013; this period witnessed Obama's reelection in the United States, the Eurozone crisis, Chinese leadership transition, and the U.S. fiscal policy

battles. Brexit in Europe and Trump's election in the United States led to the highpoints observed between 2015 and 2016. Figure 2 shows that, except for India, DEPU was also high around 2016. In Figure 3, the intensive phase of EMP in these countries is indicated by significant spikes, mostly observed around 2008, 2011–2012, and 2015.

Bootstrap Panel Granger Causality Test

Traditional causality tests require that variables are in their level form before we can estimate a vector autoregression (VAR) model for statistical inferences. In this case, we often pre-test variables to determine their stationarity form. To avoid the problem of pre-test bias, an alternative technique for testing the coefficient restrictions of level VAR model, where the underlying data series are integrated or cointegrated, is the modified Wald (MWALD) test in a lag augmented VAR (LA-VAR). It has a conventional asymptotic chi-square distribution when a VAR ($p + dmax$) is estimated, with lag order p and the maximum order of suspected integration is $dmax$. This approach was introduced by Toda and Yamamoto (1995) and gives rise to the Wald tests enhanced with the standard asymptotic chi-square distribution. The LA-VAR approach only requires the maximum order of integration of the underlying data series; therefore, the pre-test biases and size distortion problems are avoided (Yamada & Toda, 1998).

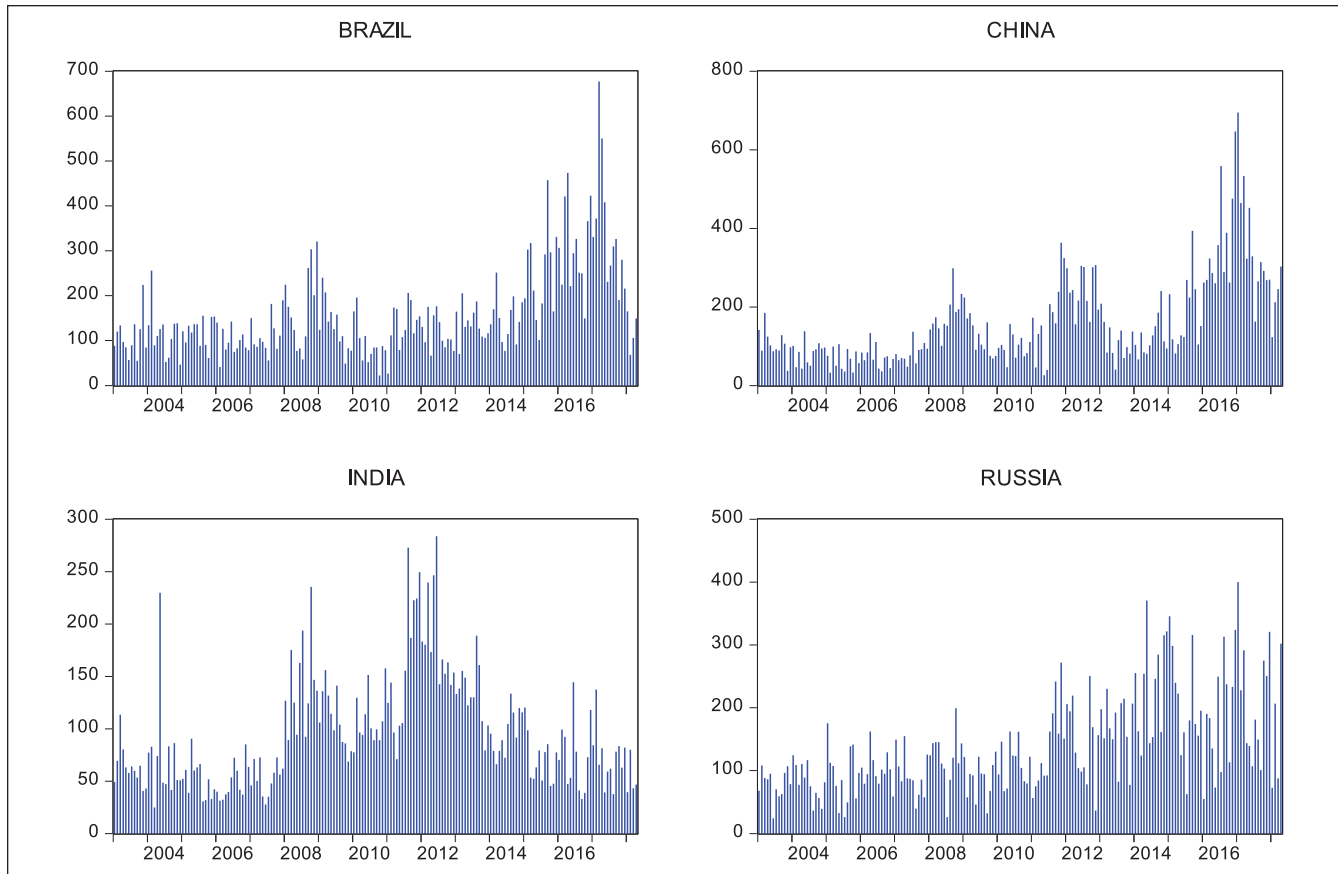


Figure 2. The DEPU index for BRIC countries.

Note. DEPU = domestic economic policy uncertainty; BRIC = Brazil, Russia, India, and China.

Emirmahmutoglu and Kose (2011) proposed a Granger causality test for heterogeneous mixed panels by extending the LA-VAR method by Fisher's (1932) meta-analysis. The regression equation for each model takes the following form:

$$\begin{aligned}
 Y_{1,t} &= \alpha_{1,1} + \sum_{i=1}^{ly_1+dmax_j} \beta_{1,1,i} Y_{1,t-i} + \sum_{i=1}^{lx_1+dmax_j} \gamma_{1,1,i} X_{1,t-i} + \varepsilon_{1,1,t} \\
 Y_{2,t} &= \alpha_{1,2} + \sum_{i=1}^{ly_1+dmax_j} \beta_{1,2,i} Y_{2,t-i} + \sum_{i=1}^{lx_1+dmax_j} \gamma_{1,2,i} X_{2,t-i} + \varepsilon_{1,2,t} \\
 &\vdots \\
 Y_{N,t} &= \alpha_{1,N} + \sum_{i=1}^{ly_1+dmax_j} \beta_{1,N,i} Y_{N,t-i} + \sum_{i=1}^{lx_1+dmax_j} \gamma_{1,N,i} X_{N,t-i} + \varepsilon_{1,N,t}
 \end{aligned} \quad (2)$$

and

$$\begin{aligned}
 X_{1,t} &= \alpha_{2,1} + \sum_{i=1}^{ly_2+dmax_j} \beta_{2,1,i} Y_{1,t-i} + \sum_{i=1}^{lx_2+dmax_j} \gamma_{2,1,i} X_{1,t-i} + \varepsilon_{2,1,t} \\
 X_{2,t} &= \alpha_{2,2} + \sum_{i=1}^{ly_2+dmax_j} \beta_{2,2,i} Y_{2,t-i} + \sum_{i=1}^{lx_2+dmax_j} \gamma_{2,2,i} X_{2,t-i} + \varepsilon_{2,2,t} \\
 &\vdots \\
 X_{N,t} &= \alpha_{2,N} + \sum_{i=1}^{ly_2+dmax_j} \beta_{2,N,i} Y_{N,t-i} + \sum_{i=1}^{lx_2+dmax_j} \gamma_{2,N,i} X_{N,t-i} + \varepsilon_{2,N,t}
 \end{aligned} \quad (3)$$

where, in the system of Equations 2 and 3, $Y_{i,t}$, $i = 1, \dots, N$ denotes EMP and $X_{i,t}$, $i = 1, \dots, N$ denotes EPU (global and domestic, respectively). N represents the number of countries included in the panel ($j = 1, \dots, N$), t refers to the time period ($t = 1, \dots, T$), l is the lag length and $dmax_j$ stands for the maximal order of integration.

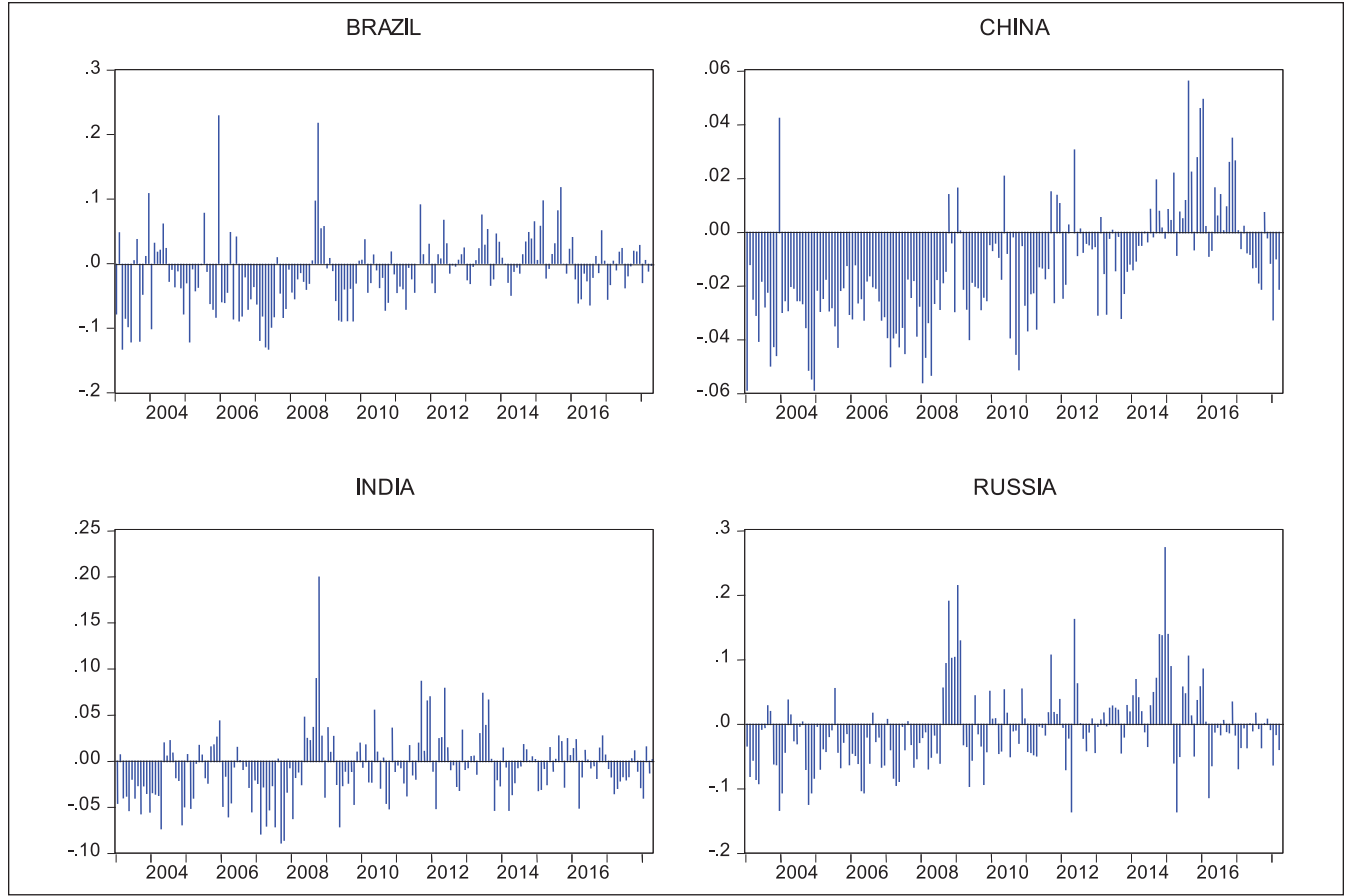


Figure 3. The EMP index for BRIC countries.

Note. BRIC = Brazil, Russia, India, and China; EMP = exchange market pressure.

From the alternative hypotheses stated earlier, the possible alternative causal relations that could be detected for each country are (a) one-way Granger causality from X (EPU) to Y (EMP) when all $\beta_{2,j}$'s are zero, but not all $\gamma_{1,j,i}$'s are zero; (b) one-way Granger causality from Y (EMP) to X (EPU) when all $\gamma_{1,j,i}$'s are zero, but not all $\beta_{2,j}$'s are zero; (c) two-way Granger causality between X and Y if some of the $\gamma_{1,j,i}$'s and $\beta_{2,j}$'s are not zero.

Employing Fisher's (1932) meta-analysis statistical procedure in Granger causality test, Emirmahmutoglu and Kose (2011) conduct separate time series tests for the individuals in the panel, obtain the significant p values from each time series test and then calculate the Fisher test statistics using the obtained p values combined in a single panel test statistic. The chi-square distribution of the panel test statistic has $2N$ degrees of freedom. The Fisher test statistic denoted as λ takes the following form:

$$\lambda = -2 \sum_{i=1}^N \ln(p_i) \quad i = 1, 2, \dots, N, \quad (4)$$

p_i denotes the p value for the Wald statistic of the i th cross section.

The limit distribution of the Fisher test statistic becomes invalid when cross-sectional dependence exists in the data series. To deal with this problem, Emirmahmutoglu and Kose (2011) test for Granger causality in the presence of cross-sectional dependence through the bootstrap approach. The simulation study shows this test is efficient even if N and T dimensions are small, whether there is cross-sectional dependency or not. However, the limitation of this test is shown in the Monte Carlo results under cross-section dependency, Emirmahmutoglu and Kose (2011) show that the LA-VAR approach could suffer from serious size distortion problems for small values of T , especially, as $N \rightarrow \infty$ without a corresponding increase in T dimension. We estimate Equations 2 and 3 and carry out panel Granger causality tests with bootstrap critical values. The steps involved in the Emirmahmutoglu and Kose (2011) bootstrap procedure for testing causality from X (GEPU and DEPU) to Y (EMP) are as follows:

Step 1. To determine the maximal order of integration ($dmax$) for each cross-section in the VAR system, Dickey and Fuller (1981) unit root tests are applied. Equation 2 is then estimated through ordinary least squares (OLS) for each cross-section and the lag orders (l_i) are selected using Schwarz Bayesian criteria, with $dmax_j$ set at 4.

Step 2. Equation 2 is re-estimated via OLS using l_i and $dmax_j$ obtained from Step 1. Then the residuals for each cross-section are extracted.

$$\hat{\varepsilon}_{1i,t}^y = Y_{i,t} - \hat{a}_{1i} - \sum_{j=1}^{l_i+dmax_j} \hat{A}_{11,ij} X_{i,t-j} - \sum_{j=1}^{l_i+dmax_j} \hat{A}_{12,ij} Y_{i,t-j} \quad (5)$$

Step 3. In line with Stine (1987), the residuals are centered as follows:

$$\tilde{\varepsilon}_{1i,t}^y = \hat{\varepsilon}_{1i,t}^y - (T-l-k-2)^{-1} \sum_{k+l+2}^T \hat{\varepsilon}_{1t} \quad (6)$$

where $\hat{\varepsilon}_{1t} = (\hat{\varepsilon}_{11t}, \hat{\varepsilon}_{12t}, \dots, \hat{\varepsilon}_{1kt})$, $l = \max(l_i)$ and $k = \max(dmax_j)$

Step 4. $[\tilde{\varepsilon}_{1i,t}^y]_{NT}$ is generated from the residuals and the bootstrap residuals are denoted as $\tilde{\varepsilon}_{1i,t}^{y*}$

where $t = 1, 2, \dots, T$.

Step 5. The bootstrap sample of Y^* is generated recursively

$$Y_{i,t}^* = \hat{a}_{1i} + \sum_{j=k_i+1}^{l_i+dmax_j} \hat{A}_{11,ij} X_{i,t-j} + \sum_{j=1}^{l_i+dmax_j} \hat{A}_{12,ij} Y_{i,t-j}^* + \tilde{\varepsilon}_{1i,t}^{y*} \quad (7)$$

Step 6. Finally, $Y_{i,t}^*$ is substituted for $Y_{i,t}$ and Equation 2 is estimated without the imposition of parameter restrictions. Wald statistics with asymptotic chi-square distribution and l_i degrees of freedom are calculated to test the null hypothesis of no causality for individual countries. Individual p values are computed from the Wald statistics, the Fisher test statistic is obtained, and bootstrap empirical distribution of the Fisher test statistics are generated by repeating all the previous 10,000 times.

To test causality from Y (EMP) to X (GEPU and DEPU), similar steps as those above are followed.

Cross-Sectional Dependence Tests

An important feature of the bootstrap panel Granger causality testing procedure is that it assumes the presence of cross-sectionally correlated errors in the data series. The implication of this in our study is that the sample countries are all affected by the impact of EPU and, therefore, manifest mutual economic characteristics. It is vital to test for cross-sectional dependence in our data series to ensure robust panel causality

results are obtained. We perform the following cross-sectional dependence tests for this purpose.

The Breusch and Pagan's (1980) Lagrange multiplier (LM) test: This tests the null hypothesis of no cross-sectional dependence against the alternative hypothesis of cross-sectional dependence and takes the following form:

$$LM = T \sum_{i=1}^{N-1} \sum_{j=i+1}^N \hat{\rho}_{ij}^2 \quad (8)$$

In Equation 8, $\hat{\rho}_{ij}^2$ refers to the sample estimate of pairwise correlation of the residuals from the estimated pooled OLS. The LM test is valid for panels with relatively small cross-sectional N dimension and sufficiently large time T dimension.

Pesaran (2004) LM test: This was proposed for cases of large panels under the null hypothesis of no cross-sectional dependence. It is in the following form:

$$CD_{lm} = \left(\frac{1}{N(N-1)} \right)^{\frac{1}{2}} \sum_{i=1}^{N-1} \sum_{j=i+1}^N (T \hat{\rho}_{ij}^2 - 1) \quad (9)$$

The CD_{lm} test will converge to a standard normal distribution under the null hypothesis. One challenge of this test is that it is subject to size distortions when N is relatively large and T is relatively small.

Pesaran (2004) cross-sectional dependency (CD) test: This is a more general test valid for panels large on both N and T dimensions, derived from the average of pairwise correlation coefficients, $\hat{\rho}_{ij}$. It has a null of no cross-sectional dependence and takes the following form:

$$CD = \sqrt{\left(\frac{2T}{N(N-1)} \right)} \left(\sum_{i=1}^{N-1} \sum_{j=i+1}^N \hat{\rho}_{ij} \right) \quad (10)$$

This CD test has an asymptotic standard normal distribution under the null hypothesis. It is robust for heterogeneous models that are characterized by multiple breaks in their slope coefficients as well as their error variances, provided the unconditional means of the variables do not vary with time and their innovations are symmetric. However, this test has limited nonzero population average pairwise correlations (Pesaran, Ullah, & Yamagata, 2008).

Pesaran et al. (2008) LM test: This is a modified version of the LM test which uses both mean and variance of LM statistic under the null hypothesis of no cross-sectional dependence. It is known as bias-adjusted LM test, and it takes the following form:

Table 2. Cross-Sectional Dependence and Slope Homogeneity Tests.

Variables	Breusch-Pagan LM	Pesaran scaled LM	Bias-corrected scaled LM	Pesaran CD	$\hat{\Delta}$	$\hat{\Delta}_{adj}$
GEPU	1104.000*** (0.000)	315.811*** (0.000)	315.799*** (0.000)	33.226*** (0.000)	19.865*** (0.000)	20.029*** (0.000)
DEPU	170.271*** (0.000)	46.266*** (0.000)	46.255*** (0.000)	10.900*** (0.000)	8.255*** (0.000)	8.323*** (0.000)
EMP	284.681*** (0.000)	79.293*** (0.000)	79.282*** (0.000)	16.835*** (0.000)	30.871*** (0.000)	31.125*** (0.000)

Note. The p values are in parentheses. LM = Lagrange multiplier; CD = cross-sectional dependency; GEPU = global economic policy uncertainty; DEPU = domestic economic policy uncertainty; EMP = exchange market pressure.

*** indicates significance at 1% significance level.

$$LM_{adj} = \sqrt{\left(\frac{2T}{N(N-1)}\right)} \sum_{i=1}^{N-1} \sum_{j=i+1}^N \hat{\rho}_{ij} \frac{(T-k)\hat{\rho}_{ij}^2 - \mu_{Tij}}{\sqrt{v_{Tij}^2}}. \quad (11)$$

In Equation 11, μ_{Tij} is the exact mean of $(T-k)\hat{\rho}_{ij}^2$ while v_{Tij}^2 is its exact variance (Pesaran et al., 2008).

Slope Homogeneity Test

Bootstrap panel causality technique also allows slope heterogeneity across countries. This important feature involves the application of the Wald principle, valid for cases of small cross-section N dimension and large time T dimension. It tests the null hypothesis of slope homogeneity of coefficient against the alternative hypothesis. Swamy (1970) further developed the slope homogeneity test to detect cross-sectional heteroscedasticity, but both Wald and Swamy tests are not applicable for all panel models data because of the size restriction. Pesaran and Yamagata (2008) introduced a standardized form of Swamy's test known as the $\tilde{\Delta}$ test. It is applicable for testing slope homogeneity without any restrictions to the expansion of N and T dimensions in panel models and it is valid as $(N, T) \rightarrow \infty$ when error terms are normally distributed. The $\tilde{\Delta}$ test requires the computation of the modified version of Swamy's test in the following form:

$$\tilde{S} = \sum_{i=1}^N \left(\hat{\beta}_i - \tilde{\beta}_{WFE} \right) \frac{x_i' M_{\tau} x_i}{\tilde{\sigma}_i^2} \left(\hat{\beta}_i - \tilde{\beta}_{WFE} \right) \quad (12)$$

where $\hat{\beta}_i$ = pooled OLS estimator, $\tilde{\beta}_{WFE}$ = weighted fixed effect pooled estimator, and $\tilde{\sigma}_i^2$ is the estimator.

Next, the standard dispersion statistics is computed as

$$\tilde{\Delta} = \sqrt{N} \left(\frac{N^{-1} \tilde{S} - k}{2k} \right). \quad (13)$$

The bias-adjusted version of the standard dispersion statistics is given as

$$\tilde{\Delta}_{Adj} = \sqrt{N} \left(\frac{N^{-1} \tilde{S} - E(\tilde{z}_{it})}{\sqrt{\text{var}(\tilde{z}_{it})}} \right). \quad (14)$$

Empirical Results

We make appropriate inferences from our estimations in this section having performed the cross-sectional dependence and the slope homogeneity tests which led to our choice of the bootstrap panel causality for our estimation.

Table 2 reports the results of cross-sectional dependence tests (LM, CD_{lm} , CD, and LM_{adj}). The null hypothesis of no cross-sectional dependence is rejected at 1% level of significance for all the four test statistics. This shows that the error terms in the data series are cross-sectionally correlated. This implies that the BRIC countries are economically integrated and highly interconnected in the exchange market; therefore, policy uncertainty affecting any of the economies will likely affect the other countries.

The Pesaran and Yamagata (2008) slope homogeneity test results for our data series are also reported in Table 2. We find evidence of country-specific heterogeneity because the two tests, $\hat{\Delta}$ and $\hat{\Delta}_{adj}$ reject the null hypothesis of the slope homogeneity in our series. This suggests that the direction of causal relations between EPU and EMP may differ across the four countries; hence, the variables do not require that we impose homogeneity restrictions, otherwise our inferences might be misleading.

Having confirmed cross-sectional dependence and heterogeneity of slopes across these four countries, the bootstrap panel causality is a convenient means to obtain reliable results. The final results from the bootstrap panel Granger causality procedure are reported in Table 3.

Results in Table 3 show significant one-way causal effects from GEPU to EMP and from DEPU to EMP in Brazil. These findings suggest that domestic uncertainty arising from Brazil's changing policies influences its EMP, and the GEPU also exerts some degree of influence on the Brazilian EMP. Therefore, Brazil should be concerned about the effects of global disturbances on its currency. Furthermore, it should be

Table 3. Granger Causality.

Countries	Lag	Wald statistics	p values	Countries	Lag	Wald statistics	p values
A. Granger causality between global economic policy uncertainty and exchange market pressure							
H_0 : EMP does not Granger cause GEPU				H_0 : GEPU does not Granger cause EMP			
Brazil	3	0.826	.843	Brazil	3	6.455	.026**
China	3	6.068	.014**	China	3	4.428	.012**
India	1	5.336	.149	India	1	3.694	.061*
Russia	3	1.758	.624	Russia	3	1.473	.416
Panel		13.666	.244	Panel		14.725	.035**
B. Granger causality between domestic economic policy uncertainty and exchange market pressure							
H_0 : EMP does not Granger cause DEPU				H_0 : DEPU does not Granger cause EMP			
Brazil	3	4.937	.176	Brazil	3	6.366	.095*
China	3	18.194	.000***	China	3	3.435	.329
India	3	13.650	.003***	India	3	10.047	.018**
Russia	2	9.032	.011**	Russia	2	7.195	.027**
Panel		39.498	.000***	Panel		22.138	.005***

Note. Lag orders were selected using Schwarz Bayesian criteria. EMP = exchange market pressure; GEPU = global economic policy uncertainty; DEPU = domestic economic policy uncertainty.

***, **, and * indicate significance at 1%, 5%, and 10% significance levels, respectively.

cautious of its homemade policies, because both global and domestic uncertainty can put pressure on the Real, but the pressure on the Real is not strong enough to cause uncertainty anywhere.

In the case of Russia, no significant causal relationship between GEPU and EMP was detected. However, significant bidirectional causality between DEPU and EMP was detected. This suggests that developments in the Russian foreign exchange market are not strong enough to induce economic uncertainty globally; furthermore, the Russian foreign exchange market is not sensitive to GEPU. Russia only has to be concerned about the influence of its domestic policies on its exchange rates and reserves, and the role of currency instability in generating uncertainty in the domestic economy.

Considering India, we find significant one-way Granger causality from GEPU to EMP. We also observe significant feedback causality between DEPU and EMP. We are, thus, able to infer that India's foreign exchange market is prone to the effect of both domestic and international EPU and tends to come under pressure in response to both events. This is a vicious circle situation for India because, as EMP intensifies during either global or domestic uncertainty, it will further generate domestic uncertainty in the country, but this has no influence on global uncertainty. Thus, India needs to be more concerned about issues of economic uncertainty within and around the world, because the negative effects will not be limited to the exchange market only; this will transmit to other economic activities through its foreign exchange. Expected policy response to these changes will further lead to DEPU.

As for China, Table 3 reports a significant bidirectional causal relationship between GEPU and EMP. It also reports a significant one-way Granger causality running from EMP to DEPU. This implies that both external policy changes and currency stabilization should be issues of concern for China. It is noteworthy that EMP in China is strong enough, not only to predict DEPU, but it also causes GEPU. On the one hand, the Chinese foreign exchange market is susceptible to the influence of international EPU spillovers and, on the other, fluctuations in the Chinese foreign exchange market are strong enough to induce GEPU. This is indicative of China's new status as a global economic power, especially as the Chinese Yuan joins the USD, Euro, Yen, and BPS in the International Monetary Fund's basket of reserve currencies. This also indicates the strong involvement of the Chinese currency in bilateral trades with other countries and their exchange rate serves as a manipulative tool which China uses to enhance its international competitiveness. However, it is expected that monetary policy might often respond to currency distortion via interest rates, as shown in the empirical evidence by Bjørnland and Halvorsen (2014); hence, EMP influences DEPU.

When we consider BRIC countries as a whole (that is, for the entire panel), causality results indicate the existence of a significant unidirectional causality from GEPU to EMP. The results also indicate the presence of a significant feedback causal relationship between DEPU and EMP. First, the results suggest that, overall, the foreign exchange markets of BRIC countries are exposed to the influence of international EPU spillovers. Second, the results show that there is a

vicious cycle in which DEPU induces pressure in the foreign exchange markets of BRIC countries and, in turn, EMP reinforces DEPU, thus implying that both mutually interact with each other. Moreover, our panel findings show more insights about the relationships GEPU, DEPU, and EMP.

The new insight we provide on the relationship between EPU and the exchange market is that we approach the exchange market by observing the joint movements that occur in the two external variables (foreign reserves and exchange rates) which indicate the EMP. Given that Aizenman and Binici (2016) have found that external factors have significant effects on EMP with larger impact on emerging market economies, in line with this, our findings have also shown that GEPU is another external factor that influences EMP. Our results are in line with the documented literature on the relationship between exchange rates and EPU; for instance, Krol (2014) documented that both home and foreign EPU increase exchange rate volatility for currencies of emerging economies. This is also similar to the results of Kido (2016) who investigated the spillover effects of international EPU on real effective exchange rates and showed that high-yielding currencies (Brazilian Real inclusive) are negatively correlated with the international EPU and the intensity of correlations tends to be higher during the high external uncertainty. Our findings on Russia are similar to results obtained by Das and Kumar (2018) who investigated the effects of DEPU on emerging markets stock prices and found that there is evidence that emerging markets are more sensitive to domestic uncertainty than to international uncertainty. We have also extended this literature by showing that EMP can significantly reinforce DEPU, and, for a large economy like China, its exchange market is important in influencing GEPU.

Policy Implications

The following implications are deduced from our overall findings:

First, GEPU causes EMP in Brazil, China, and India. This suggests that, in these countries, the future prediction of pressure in foreign exchange can be based on global economic events that generate policy uncertainties around the world. Exchange market literature has recognized uncertainty as a negative, but nonnegligible, determinant of fluctuations in exchange market rate. This implies that uncertainty that arises from abroad may have domestic consequences, and the medium of transmission is the exchange market. Therefore, a shift should be expected in the exchange market activities of Brazil, China, and India, while the magnitude of global uncertainty spillover will likely be higher for the three than for Russia. Staying aware of the harmful effects of global economic disturbances will ensure their swift defensive measures.

Second, on the domestic scene, significant causality is observed from DEPU to EMP, which implies that domestic uncertainty could be a factor that dictates the foreign EMP. Based on the established negative effect of uncertainty on

economic activities, pressures in the foreign exchange market, having domestic origin, except in China, implies that policy makers may often be forced to devalue their currencies to normalize their economies rather than putting pressure on their reserves.

Third, it is mostly observed that EMP is Granger causal to DEPU except for Brazil. This suggests that foreign exchange market instabilities can generate further domestic policy uncertainty; hence, domestic uncertainty in countries is externally induced through the exchange market. China, India, and Russia need to be concerned about the foreign exchange management to avoid the spread of further uncertainty within their economies. Brazil only needs to focus on policies that prevent domestic uncertainty, because its EMP cannot cause domestic uncertainty. It is not surprising that, apart from China, their currency instabilities are not strong enough to cause global uncertainty. The economies in partnership with China, therefore, need to be concerned when pressures hit the Chinese currency.

Fourth, the presence of cross-sectional dependence in the data series suggests that some degree of economic alliance and mutual developmental characteristics can be found among BRIC countries. This indicates that issues surrounding uncertainty and the exchange market can transcend from any of the economies to affect the other countries. However, the presence of heterogeneous slopes also shows that each of the BRIC economies retains their unique characteristics. This suggests that the direction of causal relations between EPU and EMP may differ across countries.

Finally, the role of uncertainty is magnified by the interconnectivity of these four largest emerging market economies. Although GEPU does not matter for Russia's EMP, we quite expect that BRIC market economies may experience pressures in their exchange rate during high global economic uncertainty due to their interconnection.

Conclusion

The interconnection of the world's economies has made the effects of economic disturbances more prevalent while the defensive response from each country has generated considerable EPU. This article, therefore, studies causal relationships between EMP and EPU (global and domestic) in the BRIC countries. Emirmahmutoglu and Kose (2011) bootstrap panel causality procedure was applied to two bivariate models, one which examined GEPU–EMP relationship while the second model examined DEPU–EMP relationship.

The results reveal some notable relationships of EMP with both GEPU and DEPU. First, only China has bidirectional causality running between GEPU and EMP, while no causality was found in this case for Russia. Both India and Russia have bidirectional causality running between DEPU and EMP. DEPU does not matter for China's EMP, but the EMP is Granger causal for its DEPU. Generally, a unidirectional causality runs from GEPU to EMP, while feedback

causality was found between EMP and DEPU for the panel; this implies that both variables mutually interact with each other.

Despite the varying relationships between EPU and EMP across these countries due to their different conditions, empirical evidence suggests that both global and domestic economic policy uncertainties significantly help to predict the future pressures in the exchange market of the BRIC countries. These suggest a need to monitor the issues of GEPU and DEPU closely, as well as the conditions in the exchange market during crisis. An effective management of the situation will ease the pressures that might be caused by speculative attacks on currencies.

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References

- Aizenman, J., & Binici, M. (2016). Exchange market pressure in OECD and emerging economies: Domestic vs. factors and capital flows in the old and new normal. *Journal of International Money and Finance*, 66, 65-87.
- Aizenman, J., & Hutchison, M. M. (2012). Exchange market pressure and absorption by international reserves: Emerging markets and fear of reserve loss during the 2008-2009 crisis. *Journal of International Money and Finance*, 31, 1076-1091.
- Aizenman, J., & Pasricha, G. K. (2012). Determinants of financial stress and recovery during the great recession. *International Journal of Finance & Economics*, 17, 347-372.
- Akram, G. M., & Byrne, J. P. (2015). Foreign exchange market pressure and capital controls. *Journal of International Financial Markets, Institutions & Money*, 37, 42-53.
- Arbatli, E. C., Davis, S. J., Ito, A., Miake, N., & Saito, I. (2017). *Policy uncertainty in Japan* (No. w23411). Cambridge, MA: National Bureau of Economic Research.
- Baker, S. R., Bloom, N., & Davis, S. J. (2016). Measuring economic policy uncertainty. *The Quarterly Journal of Economics*, 131, 1593-1636.
- Balcilar, M., Gupta, R., Kyei, C., & Wohar, M. E. (2016). Does economic policy uncertainty predict exchange rate returns and volatility? Evidence from a nonparametric causality-in-quantiles test. *Open Economies Review*, 27, 229-250.
- Beckmann, J., & Czudaj, R. (2017). Exchange rate expectations and economic policy uncertainty. *European Journal of Political Economy*, 47, 148-162.
- Belke, A., & Osowski, T. (2017). *International effects of Euro area versus US policy uncertainty: A FAVAR approach* (Ruhr Economic Papers No. 689). Retrieved from <https://ideas.repec.org/p/zbw/glodps/35.html>
- Benigno, G., Benigno, P., & Nisticò, S. (2012). Risk, monetary policy, and the exchange rate. *NBER Macroeconomics Annual*, 26, 247-309.
- Bernal, O., Gnabo, J. Y., & Guilmin, G. (2016). Economic policy uncertainty and risk spillovers in the Eurozone. *Journal of International Money and Finance*, 65, 24-45.
- Bird, G., & Mandilaras, A. (2006). Regional heterogeneity in the relationship between fiscal imbalances and foreign exchange market pressure. *World Development*, 34, 1171-1181.
- Bjørnland, H. C., & Halvorsen, J. I. (2014). How does monetary policy respond to exchange rate movements? New international evidence. *Oxford Bulletin of Economics and Statistics*, 76, 208-232.
- Bloom, N. (2017). Observations on uncertainty. *Australian Economic Review*, 50, 79-84.
- Bouri, E., Gupta, R., Hosseini, S., & Lau, C. K. M. (2017). Does global fear predict fear in BRICS stock markets? Evidence from a Bayesian graphical structural VAR model. *Emerging Markets Review*, 34, 124-142.
- Boyer, J., & Truman, E. M. (2005, October). The United States and the large emerging-market economies: Competitors or partners. The United States and the world economy: Foreign economic policy for the next decade. *Foreign Affairs*, 143-172.
- Breusch, T. S., & Pagan, A. R. (1980). The Lagrange multiplier test and its applications to model specification in econometrics. *The Review of Economic Studies*, 47, 239-253.
- Calvo, G. A., & Reinhart, C. M. (2002). Fear of floating. *The Quarterly Journal of Economics*, 117, 379-408.
- Colombo, V. (2013). Economic policy uncertainty in the US: Does it matter for the Euro area? *Economics Letters*, 121, 39-42.
- Dakhlaoui, I., & Aloui, C. (2016). The interactive relationship between the US economic policy uncertainty and BRIC stock markets. *International Economics*, 146, 141-157.
- Das, D., & Kumar, S. B. (2018). International economic policy uncertainty and stock prices revisited: Multiple and partial wavelet approach. *Economics Letters*, 164, 100-108.
- Davis, S. J. (2016). *An index of global economic policy uncertainty* (No. W22740). Cambridge, MA: National Bureau of Economic Research.
- Dickey, D. A., & Fuller, W. A. (1981). Likelihood ratio statistics for autoregressive time series with a unit root. *Econometrica: Journal of the Econometric Society*, 1057-1072.
- Eichengreen, B., Rose, A. K., & Wyplosz, C. (1996). *Contagious currency crises* (No. W5681). Cambridge, MA: National Bureau of Economic Research.
- Emirmahmutoglu, F., & Kose, N. (2011). Testing for Granger causality in heterogeneous mixed panels. *Economic Modelling*, 28, 870-876.
- Feldkircher, M., Horvath, R., & Rusnak, M. (2014). Exchange market pressures during the financial crisis: A Bayesian model averaging evidence. *Journal of International Money and Finance*, 40, 21-41.
- Fisher, R. A. (1932). *Statistical methods for research workers* (4th ed.). Edinburgh, Scotland: Oliver and Boyd.
- Georgiadis, G. (2016). Determinants of global spillovers from US monetary policy. *Journal of International Money and Finance*, 67, 41-61.

- Girton, L., & Roper, D. (1977). A monetary model of exchange market pressure applied to the postwar Canadian experience. *The American Economic Review*, 537-548.
- Gochoco-Bautista, M. S., & Bautista, C. C. (2005). Monetary policy and exchange market pressure: The case of the Philippines. *Journal of Macroeconomics*, 27, 153-168.
- Hallwood, C. P., & Marsh, I. W. (2004). Exchange market pressure on the pound-dollar exchange rate: 1925-1931. *The North American Journal of Economics and Finance*, 15, 249-264.
- Handley, K. (2014). Exporting under trade policy uncertainty: Theory and evidence. *Journal of International Economics*, 94, 50-66.
- Hegerty, S. W. (2012). Money market pressure in emerging economies: International contagion versus domestic determinants. *Economic Systems*, 36, 506-521.
- International Monetary Fund. (2013). *World economic outlook: Hopes, realities risks*. IMF Press. Retrieved from <http://www.imf.org/external/pubs/ft/weo/2013/01/>
- International Monetary Fund. (2014). *World economic outlook: Legacies, clouds, uncertainties*. IMF Press. Retrieved from <http://www.imf.org/external/pubs/ft/weo/2014/02/>
- Kang, W., Lee, K., & Ratti, R. A. (2014). Economic policy uncertainty and firm-level investment. *Journal of Macroeconomics*, 39, 42-53.
- Kang, W., & Ratti, R. A. (2013). Oil shocks, policy uncertainty and stock market return. *Journal of International Financial Markets, Institutions and Money*, 26, 305-318.
- Kido, Y. (2016). On the link between the US economic policy uncertainty and exchange rates. *Economics Letters*, 144, 49-52.
- Ko, J. H., & Lee, C. M. (2015). International economic policy uncertainty and stock prices: Wavelet approach. *Economics Letters*, 134, 118-122.
- Krol, R. (2014). Economic policy uncertainty and exchange rate volatility. *International Finance*, 17, 241-256.
- Leippold, M., & Matthys, F. (2015). *Economic policy uncertainty and the yield curve*. Retrieved from https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2669500
- Luk, P., Cheng, M., Ng, P., & Wong, K. (2017). Economic policy uncertainty spillovers in small open economies: The case of Hong Kong. *Pacific Economic Review*. Advance online publication. doi:10.1111/1468-0106.12283
- Maradiaga, D. I., Zapata, H. O., & Pujula, A. L. (2012, 4-7 February). Exchange rate volatility in BRICS countries. In *Southern Agricultural Economics Association, Annual Meeting* (pp. 4-7). Birmingham, Alabama (No.119726).
- Panday, A. (2015). Impact of monetary policy on exchange market pressure: The case of Nepal. *Journal of Asian Economics*, 37, 59-71.
- Pesaran, M. H. (2004). *General diagnostic tests for cross section dependence in panels* (Cesifo Working Paper Series No. 1229, IZA Discussion Paper No. 1240). Retrieved from <http://Ssrn.Com/Abstract=572504>
- Pesaran, M. H., Ullah, A., & Yamagata, T. (2008). A bias-adjusted LM test of error cross-section independence. *The Econometrics Journal*, 11, 105-127.
- Pesaran, M. H., & Yamagata, T. (2008). Testing slope homogeneity in large panels. *Journal of Econometrics*, 142, 50-93.
- Sarwar, G. (2012). Is VIX an investor fear gauge in BRIC equity markets? *Journal of Multinational Financial Management*, 22, 55-65.
- Stine, R. A. (1987). Estimating properties of autoregressive forecasts. *Journal of the American Statistical Association*, 82, 1072-1078.
- Swamy, P. A. (1970). Efficient inference in a random coefficient regression model. *Econometrica: Journal of the Econometric Society*, 38, 311-323.
- Toda, H. Y., & Yamamoto, T. (1995). Statistical inference in vector autoregressions with possibly integrated processes. *Journal of Econometrics*, 66, 225-250.
- Weymark, D. N. (1995). Estimating exchange market pressure and the degree of exchange market intervention for Canada. *Journal of International Economics*, 39, 273-295.
- Weymark, D. N. (1997). Measuring the degree of exchange market intervention in a small open economy. *Journal of International Money and Finance*, 16, 55-79.
- Weymark, D. N. (1998). A general approach to measuring exchange market pressure. *Oxford Economic Papers*, 50, 106-121.
- Wilson, D., & Purushothaman, R. (2003). Dreaming with BRICs: The path to 2050. In S. C. Jain (Ed.), *Emerging economies and the transformation of international business: Brazil, Russia, India, China* (pp. 1-24). (Global Economics Paper No. 99).
- Yamada, H., & Toda, H. Y. (1998). Inference in possibly integrated vector autoregressive models: Some finite sample evidence. *Journal of Econometrics*, 86, 55-95.
- Yu, H., Fang, L., & Sun, B. (2018). The role of global economic policy uncertainty in long-run volatilities and correlations of US industry-level stock returns and crude oil. *PLoS ONE*, 13(2), E0192305.

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